2.2. FUNDAMENTAL QUANTITIES FOR ANALYSIS

There are two fundamental quantities for electrical engineering :

(*i*) Charge (*ii*) Energy.

(*i*) **Charge.** As discussed in Chapter 1, effect of charge was first observed in 600 B.C. by Greeks. But it was first defined by the French Army colonel Charles Coulomb in the form of Coulomb's inverse square law of force. This law was similar to the inverse square law proposed by Newton for force between two masses. According to Coulomb's law :

"Force of attraction or repulsion between two point charges is directly proportional to the product of charges and inversely proportional to square of distance between them"

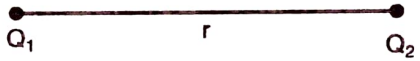


Fig. 2.1.

Q_1, Q_2 are two charges in coulomb, r is the distance in meter

then, $$F \propto \frac{Q_1 Q_2}{r^2} \text{ in Newton}$$

$$F = \frac{1}{4 \pi \varepsilon_0} \frac{Q_1 Q_2}{r^2} \qquad ...(2.1)$$

where $1/4 \pi \varepsilon_0$ is the constant in SI unit.

ε_0 is called the permittivity of free space. Its value is 8.85×10^{-12} Farad/meter.

The unit of charge, coulomb (C) is defined as the magnitude of charge, which when placed at the distance of one meter from the equal charge in free space, repels it with the force of $\frac{1}{4 \pi \varepsilon_0}$ Newtons.

The charge on the electron or proton is considered as the basic charge. Electron has the negative charge of magnitude 1.6×10^{-19} coulomb, while the proton has the same charge of positive sign.

(*ii*) **Energy.** Energy is the capacity of doing any work. The unit of energy is Joule. The most Fundamental law is the law of conservation of energy. According to this law :

"Energy cannot be created nor be destroyed, it can merely be converted from one form to another".

As we know that the electrical energy is not available in nature, so it must be generated from the other natural forms of energy. Electrical energy may be generated from chemical energy (in batteries), from light energy (in photovoltaic cell), from mechanical energy (in generators) or from nuclear fuel.

Electrical energy is the most efficient form among the all other forms of energy. It can be generated centrally in bulk and transmitted economically over long distances. Further, it can be adopted easily and efficiently to domestic and industrial applications.

2.3. DERIVED QUANTITIES FROM THE FUNDAMENTAL QUANTITIES

(*i*) **Current.** Electric current is defined as the time rate of net motion of electric charge in any conductor. It's unit is ampere,

$$i = \frac{dq}{dt} \text{ Ampere.} \qquad \ldots(2.2)$$

(*ii*) **Potential.** Potential is defined as the work done to transfer unit positive charge from infinity to any point in electric field. It's unit is volt.

$$v = \frac{dw}{dq} \text{ volt or joules/coulomb.} \qquad \ldots(2.3)$$

It is also known as the voltage. The voltage of an energy source is known as the electromotive force or emf. Emf is defined as the total work done to move an unit positive charge in any closed path of a circuit.

(*iii*) **Potential difference.** It is given by the work required to transfer unit positive charge from one point to the other. Electrical energy (or current) flows from higher potential to lower potential. So for existance of current potential difference is necessary. Emf is measured by the potential difference between the terminals when the generator (or battery) is not delivering current.

(*iv*) **Power.** As we know that power is defined as the time rate of change of energy. Power is give by p.

$$p = \frac{dW}{dt} = \frac{dW}{dq} \times \frac{dq}{dt} = v \times i \qquad ...(2.4)$$

Energy is given by, $W = \int P\, dt$ joules ...(2.5)

2.4. CIRCUIT ELEMENTS

Circuit elements may be classified in following ways :

1. Active and passive elements :

Active elements. These are the energy suppliers in electrical network. All the energy sources come in this category.

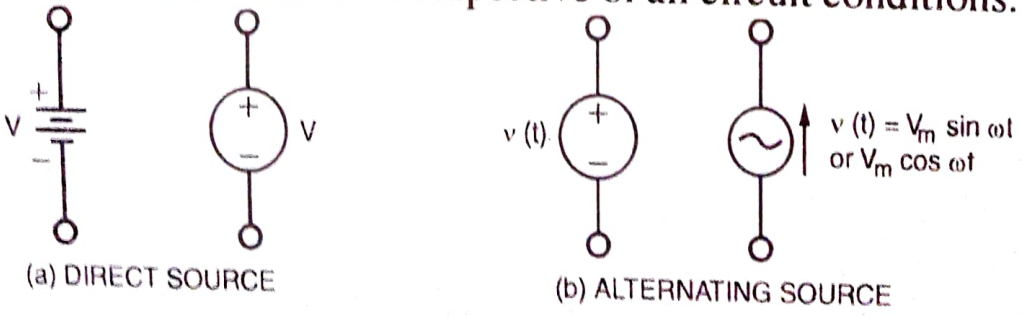
Active elements may be classified as :

(*i*) Voltage source (*ii*) Current source.

These sources may be independent or dependent.

(*i*) **Independent sources.**

(*a*) **Independent voltage source.** It is the two terminal device. The voltage across these two terminals remain constant irrespective of all circuit conditions.



(a) DIRECT SOURCE (b) ALTERNATING SOURCE

Fig. 2.2. Symbolic representation of Voltage Source.

Fig. 2.2. Symbolic representation of Voltage Source

Symbol for Independent source :

Ideal voltage source. Independent voltage source, which has zero internal resistance is considered as the ideal source. The V-I characteristic of ideal voltage source in shown in Fig. 2.3.

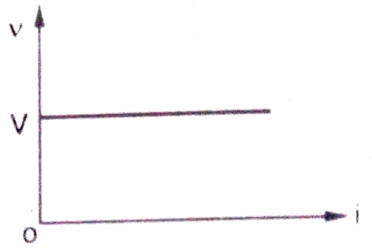
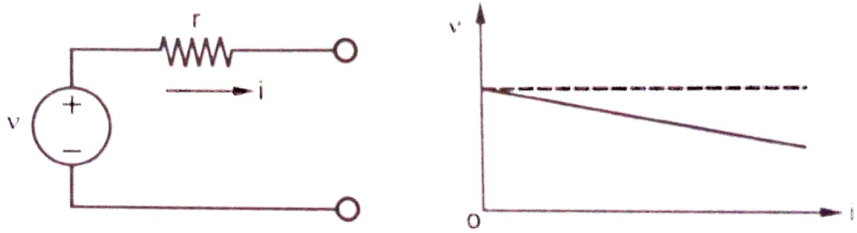


Fig. 2.3. V-I Characteristic for ideal voltage source.

Practical voltage source. Source having some amount of internal resistance is known as the Practical source. Due to this resistance, voltage drop takes place and it causes the terminal voltage to reduce. The V-I characteristics alongwith the practical source is shown in Fig. 2.4.



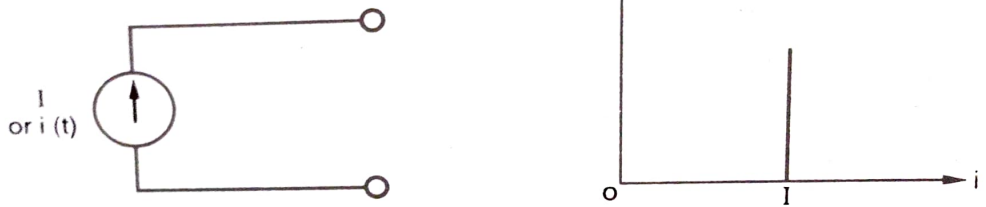
(i) PRACTICAL VOLTAGE SOURCE.

(ii) V - I CHARACTERISTIC.

Fig. 2.4.

(*b*) **Independent current source.** It is also two terminal device, and gives the constant current, irrespective the external circuit condition.

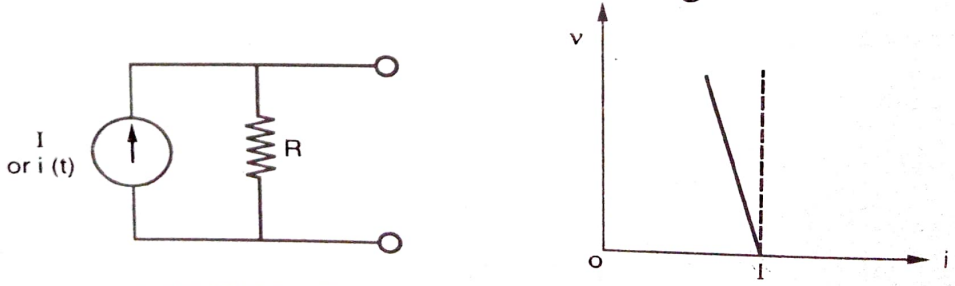
Ideal current source. Ideal current source has the infinite resistance. The source symbol alongwith its V-I characteristic is shown in Fig. 2.5.



(ii) IDEAL CURRENT SOURCE (ii) V - I CHARACTERISTIC

Fig. 2.5.

Practical current source. It has the internal resistance R (in Mega ohms). The source symbol and its V-I characteristic are shown in Fig. 2.6.



(a) SOURCE SYMBOL (b) V - I CHARACTERISTIC

Fig. 2.6. Practical Current Source.

Fig. 2.6. Practical

(*ii*) **Dependent (or controlled) energy sources.** These are four terminal devices and classified as :

(*a*) **Voltage controlled voltage source (VCVS).** Symbol for VCVS is shown in Fig. 2.7. Here the voltage across the source terminal V_{ab} is dependent upon the V_{cd}, that is the voltage across the any element in the network.

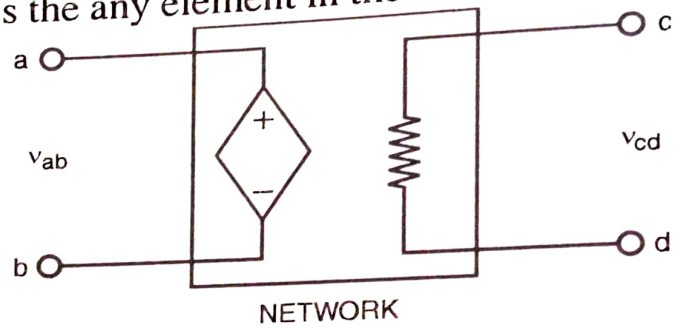


Fig. 2.7. Voltage Controlled Voltage Source (VCVS).

$$v_{ab} \propto v_{cd}$$

$$v_{ab} = k v_{cd}$$

where k = Constant (dimensionless).

(*b*) **Voltage controlled current source (VCCS).** In this dependent source current of the source, i_{ab} depends on the voltage v_{cd} . (Fig. 2.8).

$$i_{ab} \propto v_{cd} \qquad\qquad i_{ab} = \eta v_{cd}$$

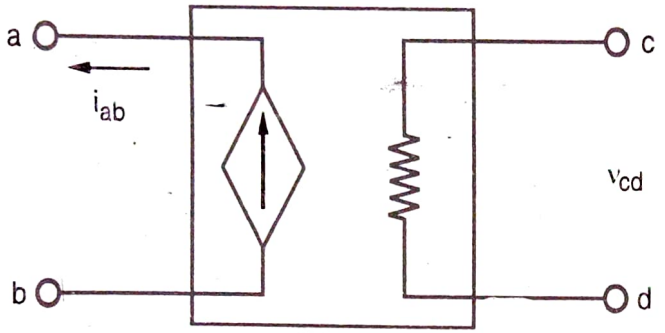


Fig. 2.8. VCCS.

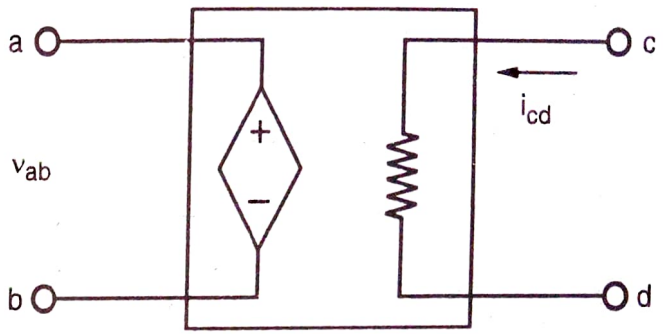


Fig. 2.9. CCVS.

where η is constant known as the transconductance and having the unit mho.

(*c*) **Current controlled voltage source (CCVS).** Voltage of the source v_{ab} depends on the current of branch *cd*. (Fig. 2.9).

$$v_{ab} \propto i_{cd} \qquad v_{ab} = r\, i_{cd}$$

where r is the constant, has the unit of resistance (*i.e.*, Ω).

(*d*) **Current controlled current source (CCCS).** Current of the dependent source, depends on the current of branch *cd*. (Fig. 2.10).

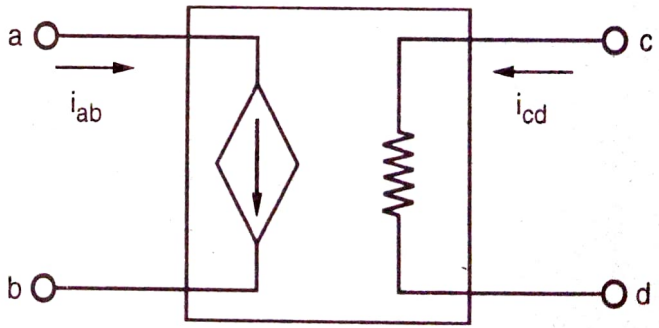


Fig. 2.10. CCCS.

$$i_{ab} \propto i_{cd}$$

$$i_{ab} = \beta\, i_{cd}$$

where β is the dimensionless constant.

Note. All semi-conductor devices like transistors, operational amplifiers and vacuum tubes are also considered as the active elements.

tubes are also considered

Passive elements. Passive elements are consumers of electrical energy. There are three passive elements in AC circuits :

(*i*) Resistance (*ii*) Inductance (*iii*) Capacitance.

(*i*) **Resistance.** Resistance is the circuit constant which was introduced in 1826 by Ohm. According to Ohm's law, the voltage across any branch is directly proportional to the current passing through it.

$$v \propto i = R\, i \qquad ...(2.6)$$

where R is the proportionallity constant known as the resistance.

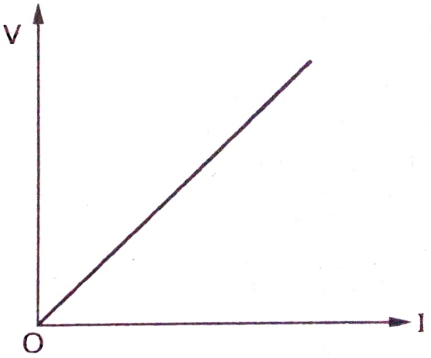


Fig. 2.11. V-I Characteristic for Resistance (Ohm's law).

Its unit is ohm (Ω). Resistance is defined as, ''the property of material due to which it opposes the flow of current through it''. The V-I characteristic for resistance is shown in Fig. 2.11, when the temperature is constant.

Resistance law. Resistance of any material is directly proportional to the length of material and inversely proportional to its area of cross-section.

$$R \propto l/A$$

$$R = \rho\, l/A \qquad ...(2.7)$$

where l = length in meter

A = Area of cross section in m^2

ρ = Resistivity or specific resistance in $\Omega - m$.

Energy dissipated in resistor. Whatever electrical energy is given to the resistor, dissipated in the form of heat.

Energy, $$W_R = \int_{-\infty}^{t} vi \; dt \text{ joules}$$

Average energy dissipated, $W_R = I^2 Rt$ joules $= \frac{V^2}{R} t$ ...(2.8)

where t = time.

Resistances in series :

$$R_{eq} = R_1 + R_2 + R_3 + \ldots$$ ...(2.9)

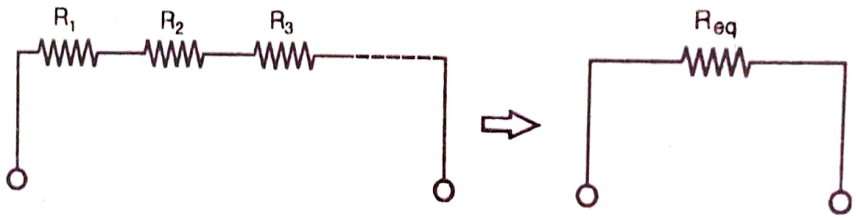


Fig. 2.12. Series Resistances.

Resistances in parallel :

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots$$ ...(2.10)

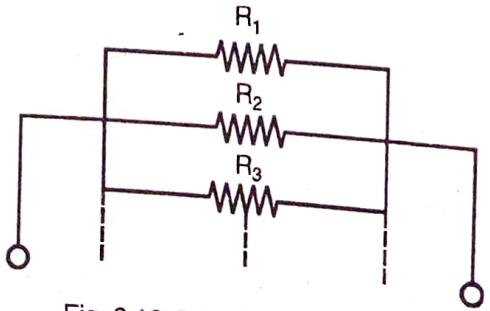


Fig. 2.13. Parallel Resistances

Fig. 2.13. Parallel Resistances.

Conductance and Conductivity. Reciprocal of resistance is known as the conductance.

It is denoted by G and its unit is mho (℧) or siemens (S)

$$G = \frac{1}{R} \qquad ...(2.11)$$

Similarly, reciprocal of resistivity (or specific resistance) is known as the conductivity. It is denoted by σ and its unit is siemens per meter.

$$\sigma = \frac{1}{\rho} \qquad ...(2.12)$$

(*ii*) **Inductance.** In the year 1820, Oersted performed the experiment to show the magnetic effect of electric current. According to Oersted, electric current, when passes through any network, produces the magnetic flux.

The total flux linkage ψ (in weber) is directly proportional to the current, i

$$\psi \propto i$$

$$= Li \qquad ...(2.13)$$

where L is the proportionality constant and known as the self inductance of the circuit. Its unit is *henry*.

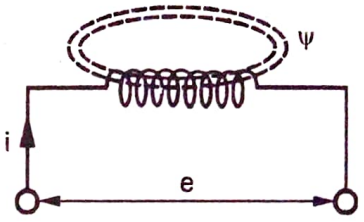


Fig. 2.14. Flux linked with inductor.

In 1831, Faraday gave the law of electromagnetic induction. According to which induced emf is directly proportional to the rate of charge of flux.

$$e \propto \frac{d\psi}{dt} \qquad ...(2.14)$$

In SI unit the proportionality constant is one, then

$$e = \frac{d\psi}{dt}$$

But, $$\psi = Li \qquad ...(2.15)$$

So the voltage across the inductor is given by

$$v_L = \frac{d\psi}{dt} = \frac{d\,(Li)}{dt} = L\frac{di}{dt} \qquad ...(2.16)$$

Current through the inductor is given by

$$i = \frac{1}{L}\int v_L\, dt. \qquad ...(2.17)$$

Now we define the inductance. "Inductance is the property of material due to which it opposes any change in current passing through it".

Similarly, *mutual inductance* can be defined between two circuits (or coils) which are closely spaced. Mutual inductance is given by M and its unit is henry.

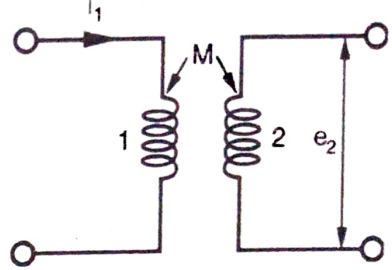


Fig. 2.15. Mutual Inductance.

Flux linked with coil 2 when current is flowing in coil 1, is given by

$$\psi_{21} = M_{21}\, i_1 \qquad ...(2.18)$$

and induced emf

$$e_2 = M_{21} \frac{di_1}{dt} \qquad ...(2.19)$$

Energy in Inductor. A pure inductor does not dissipate energy, but stores it in the form of magnetic field. Power in inductor when current i passes through it,

$$P = v\, i = L \frac{di}{dt}\, i \qquad ...(2.20)$$

So energy stored in inductor is given by

$$W = \int P\, dt = \int L \frac{di}{dt}\, i\, dt = \frac{1}{2}\, Li^2 \text{ joules.} \qquad ...(2.21)$$

Inductors in series.

$$L_{eq} = L_1 + L_2 + L_3 + \ldots \qquad \ldots(2.22)$$

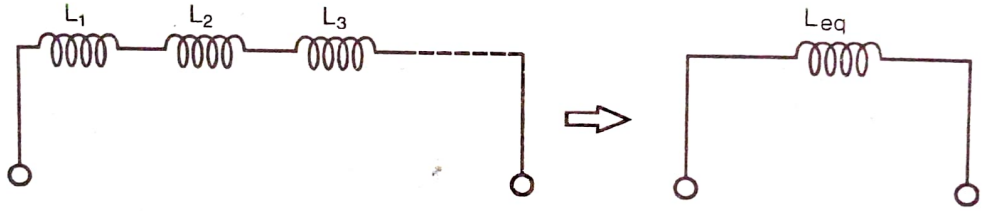


Fig. 2.16. Series Inductance.

Inductors in parallel.

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \ldots \qquad \ldots(2.23)$$

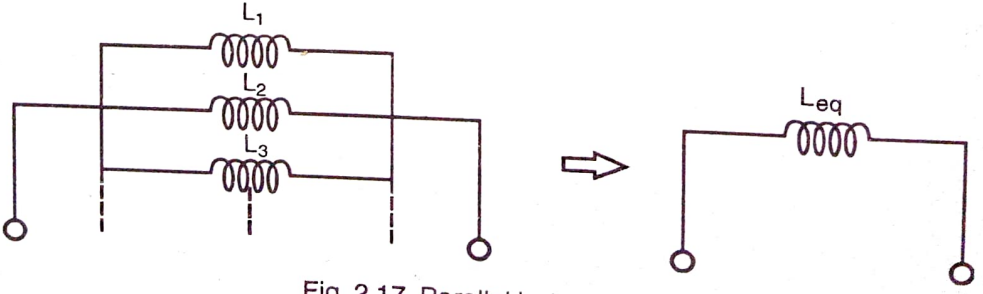


Fig. 2.17. Parallel Inductance



Fig. 2.17. Parallel Inductance.

(*iii*) **Capacitance.** It is the capability of an element to store electric charge within it. It is the property of element due to which it opposes any change in voltage across it.

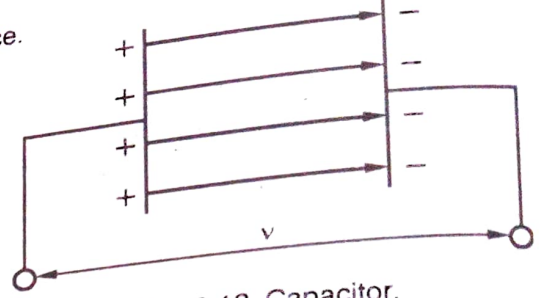


Fig. 2.18. Capacitor.

Charge stored on the capacitor plates is directly proportional to the voltage across it,

$$Q \propto v$$

$$Q = Cv \qquad \ldots(2.24)$$

where C is the constant, known as capacitance.

Its unit is Farad.

Current through the capacitor,

$$i = \frac{dQ}{dt} = \frac{d}{dt}(Cv) = C\frac{dv}{dt} \qquad \ldots(2.25)$$

Voltage across the capacitor,

$$v = \frac{1}{C}\int_{-\infty}^{t} i\ dt. \qquad \ldots(2.26)$$

Energy in capacitor. A capacitor stores electric energy in the form of electric field being established by the two polarities of charges on the two electrodes of a capacitor.

Power, $$P = v\,i = v\,C\,\frac{dv}{dt}$$

Energy stored, $$W = \int P\,dt = \int v\,C\frac{dv}{dt}\,dt = \frac{1}{2}\,C\,v^2 \text{ joules}$$ ...(2.27)

A capacitor never dissipates energy and only stores it. It can store finite amount energy, even if the current through it is zero.

Capacitances in series.

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots$$ ...(2.28)

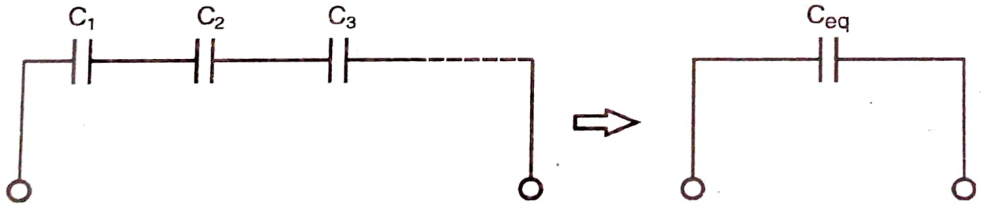


Fig. 2.19. Series Capacitors.

Capacitances in parallel.

$$C_{eq} = C_1 + C_2 + C_3 + \ldots$$ ...(2.29)

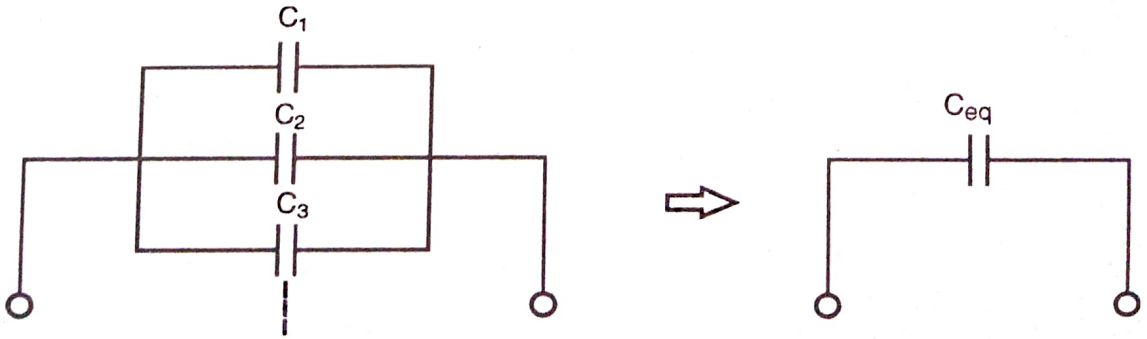
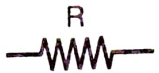
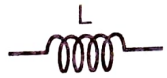


Fig. 2.20. Parallel Capacitors.

Characteristics of R, L, C are given in Table 2.1.

Table 2.1.

	Resistance	**Inductance**	**Capacitance**
Symbol			
Unit	Ohm (Ω)	Henry (H)	Farad (F)
Related law	Ohm's law $v \propto i = R\,i$	$\psi \propto i$ $= L\,i$	$Q \propto v$ $= C\,v$
Definition	It opposes the flow of current	Opposes any change in current through it	Opposes any change in voltage across it
Voltage across	$v_R = R\,i$	$v_L = L\dfrac{di}{dt}$	$v_C = \dfrac{1}{C}\int i\,dt$
Current through	$i_R = \dfrac{V}{R}$	$i_L = \dfrac{1}{L}\int v\,dt$	$i_C = C\dfrac{dv}{dt}$

	Resistance	Inductance	Capacitance
Energy	Dissipated in the form of heat, $W_R = i^2 R\, t$ joules	Stored in the magnetic form, $W_L = \frac{1}{2} L\, i^2$ joules	Stored in the electric form, $W_C = \frac{1}{2} C\, v^2$ joules
Property	Current or voltage across it can be changed instantaneously	Current across it cannot change instantaneously	Voltage across it cannot change instantaneously

2. Linear and non-linear elements. *Linear elements* are those, whose *V–I* characteristic is linear, *i.e.*, elements which obey the Ohm's law are called *linear elements.*

In general, elements obey the superposition principle are called *linear.* Resistors inductors and capacitors are linear for limited range of working. These elements are considered linear if resistance, inductance and capacitance remain constant in any active circuit.

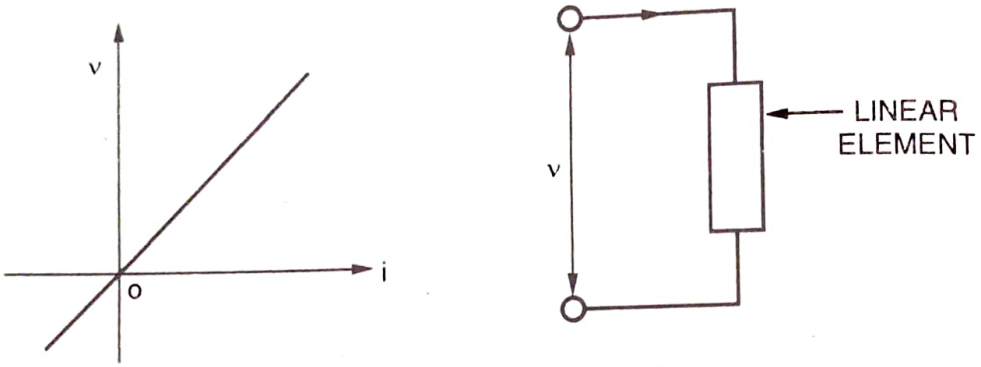


Fig. 2.21. V-I Characteristics for Linear Elements.

Non-linear elements are those, for which *V–I* characteristic is not linear or they do not obey the Ohm's law. Their values do not remain constant with time.

3. Bilateral and unilateral elements. The characteristics of bilateral elements do not depends on the direction of operation *i.e.*, if the polarity of applied voltage across them is changed, the property or the current passing through the bilateral element does not change.

Example. Resistance, inductance or capacitance.

Unilateral elements are those, which offer varying properties with variation in flow of current (or with the change in polarity of applied voltage).

Example of unilateral elements are diode, transistor etc. For one direction of current, if diode is forward biased (zero impedance), it will be reverse biased (infinite impedance) for other direction of current.

4. Lumped and distributed elements. Lumped means concentrated at a point. Elements in most of the circuits are distributed throughout the length but for

2.5. BASIC TERMS RELATED TO ELECTRIC CIRCUITS

Before stating the laws of electric circuit it will be better to understand the following terms.

(*i*) **Simple node.** Simple node is the junction where any two circuit elements are connected. In Fig. 2.22, nodes 1 and 3 are the simple nodes.

(*ii*) **Principal node.** Principal node is that junction where more than two elements are connected. Current is generally divided over here. In Fig. 2.22, nodes 2 and 4 are the principal nodes.

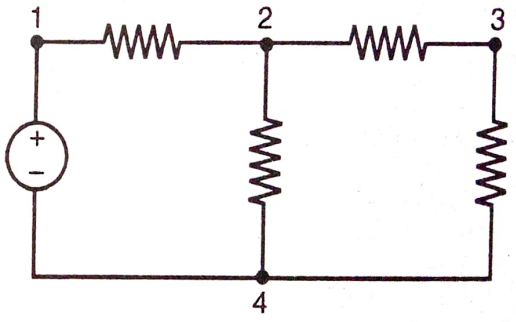


Fig. 2.22.

(*iii*) **Branch.** It is the section between two nodes in the circuit.

(*iv*) **Loop.** That is closed path made by branches in which current can flow. One loop may consist of number of meshes.

(*v*) **Mesh.** It is the loop contains no other loop with in it.

(v) Mesh. It is the loop

2.6. FUNDAMENTAL LAWS

Analysis of network is based on certain laws and rules. As said earlier in Chapter 1, that the first experimental law was presented by Coulomb in 1785, which is known as the Coulomb's inverse square law for force. After that, series of experiments were started in the field of electrical engineering by well known scientists and physicists like Ohm, Kirchhoff, Faraday and Henry. In this chapter we shall study some fundamental laws and rules on the basis of which circuit analysis will be performed in further chapters.

2.6.1. Ohm's Law. This is the most fundamental law which was given by Ohm in 1825–26. According to this law, voltage in directly proportional to current across any element when the physical conditions remain constant.

For D.C.

$$V \propto I$$

$$V = R\,I \qquad \ldots(2.30)$$

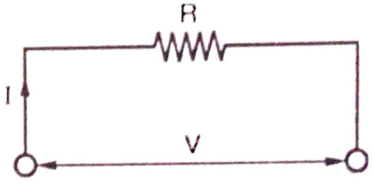


Fig. 2.23. D.C. Circuit.

where R = Resistance in ohm

For A.C,

$$v \propto i$$

$$v = Z\,i \qquad \ldots(2.31)$$

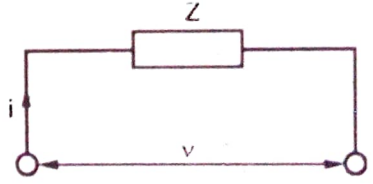


Fig. 2.24. A.C. Circuit.

Z = Impedance in ohm = $R + j\,X$

where R = Resistance in ohm

X = Reactance in ohm (inductive or capacitive).

Value of resistance varies with the variation in temperature and then Ohm's law does not hold good in any electric circuit.

So according to Ohm's law the voltage-current relationship across a resistance (or impedance) must be linear (see, Fig. 2.21).

2.6.2. Kirchhoff's Laws. In the year 1845 Kirchhoff at the age of 23, published his paper regarding two basic laws for solving the electric circuit. These laws became basis for the development of the network analysis or circuit theory.

(*i*) Kirchhoff's Current Law (KCL). It states that algebraic or vector summation of all the currents meeting at a junction or node is zero.

Or

Currents entering to the node must be equal to currents leaving the node.

According to KCL in Fig. 2.25,

$$\sum i_{\text{entering}} = \sum i_{\text{leaving}}$$

$$i_1 + i_2 + i_3 = i_4 + i_5 \qquad \ldots(2.32)$$

or $$i_4 + i_5 - i_1 - i_2 - i_3 = 0$$

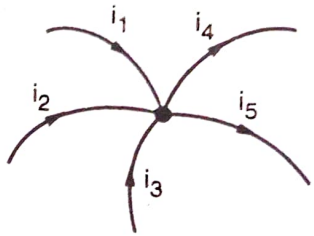


Fig. 2.25.

incoming currents are taking as negative and outgoing currents are considered positive.

Kirchhoff's current law is based on law of *Conservation of charge.*

(*ii*) **Kirchhoff's Voltage Law (KVL).** Kirchhoff's voltage law states that the total potential rise in any closed path is equal to total potential drop in that path.

Or

Algebraic or vector sum of voltages around any closed path in a circuit is always zero.

Kirchhoff's voltage law is based on law of *conservation of energy*. Since the total energy supplied to the circuit is equal to the total energy consumed in the circuit at that instant, so energy conserved in the closed path.

Energy supplied = Energy consumed.

Fig. 2.26 shows a single loop circuit with one source and two resistances R_1 and R_2.

According to KVL,

$$\Sigma \text{ Potential rise} = \Sigma \text{ Potential drop}$$

$$V = R_1 i + R_2 i \qquad ...(2.33)$$

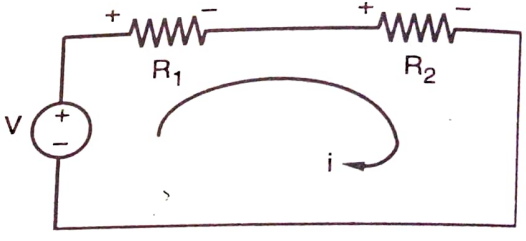


Fig. 2.26. Single loop DC circuit.

Note. According to convention current starts from the positive sign of source and flows towards the negative sign of source. (This is the positive direction of current). The polarity of potential drops is opposite to the polarity of source.

Now, consider the single loop A.C. circuit with two alternating voltage sources. one resistor and one inductor.

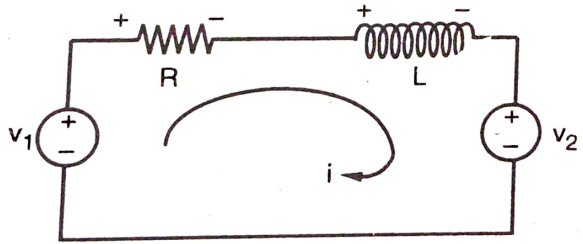


Fig. 2.27. Single loop AC circuit.

According to KVL :

$$v_1 - v_2 = Ri + L\frac{di}{dt} \qquad \text{...(2.34)}$$